Parkinson’s Freezing of Gait (FoG) Early Prediction System

Project Overview

This project presents a machine learning system for the early prediction of Parkinson’s Disease Freezing of Gait (FoG) events using wearable sensor data. The system is designed to predict FoG occurrences up to two seconds in advance by leveraging tri-axial accelerometer time series signals collected from patients during movement. The entire pipeline is implemented from scratch, covering data preprocessing, sample construction, feature engineering, model training, and evaluation, with the goal of building a complete and interpretable end-to-end machine learning workflow for time series classification in healthcare applications.

Data Processing and Sample Construction

To transform raw continuous motion signals into a supervised learning problem, the project employs a sliding window segmentation strategy combined with label forward expansion. Specifically, FoG labels are propagated backward in time to capture pre-freezing motion patterns, allowing the model to learn early warning signals before the actual event occurs. This label expansion mechanism is implemented programmatically to extend the influence of each FoG event across preceding timesteps, improving the model’s ability to detect subtle pre-event changes . The sliding window approach then segments the time series into fixed-length overlapping samples, each assigned a binary label based on whether a FoG event occurs within the window.

Feature Engineering

The project extracts a set of statistical features from each segmented time window to represent motion dynamics in a compact and informative manner. These features include mean, standard deviation, maximum, minimum, signal energy, and aggregated motion magnitude across the vertical (AccV), mediolateral (AccML), and anteroposterior (AccAP) axes. By summarizing both distributional properties and intensity of movement, the feature set captures abnormal motion patterns that typically precede FoG events. The implementation ensures consistent feature extraction across all segments, enabling effective downstream model training .

Handling Class Imbalance

Since FoG events are relatively rare compared to normal walking patterns, the dataset exhibits significant class imbalance. To address this challenge, the project incorporates imbalance handling techniques such as class weighting during model training, which increases the importance of minority class samples. This strategy is particularly effective in improving the model's sensitivity to FoG events while maintaining robustness against the dominant non-FoG class. In addition, the data construction strategy itself contributes to partially mitigating imbalance by expanding labeled regions around FoG events.

Model Implementation and Comparison

Multiple classification models are implemented and systematically compared to evaluate their performance on the FoG prediction task. These include Logistic Regression implemented from scratch using gradient descent, a Random Forest classifier leveraging ensemble learning, and a fully custom-built Neural Network with multiple fully connected layers, ReLU and sigmoid activations, and backpropagation-based optimization. The neural network architecture is designed to capture nonlinear relationships in the feature space, while the Random Forest model provides strong baseline performance through decision tree aggregation. All models are trained and evaluated using a consistent pipeline, enabling fair comparison across different algorithms .

Training and Evaluation

The dataset is split into training and testing subsets using stratified sampling to preserve class distribution. Model performance is evaluated using standard classification metrics, including precision, recall, and F1-score, with particular emphasis on recall due to the critical importance of detecting FoG events in real-world applications. The training process includes iterative optimization for neural networks and convergence-based updates for logistic regression, while the Random Forest model is trained using balanced class weights to improve minority class detection .

Results and Ongoing Improvements

The system achieves a precision of 100% and a recall of 45% on the test set, indicating that while the model is highly reliable when predicting FoG events, there is still room for improvement in capturing all occurrences. Current efforts focus on enhancing recall through more advanced feature engineering, improved sampling strategies, and hyperparameter tuning. Future work may also explore deep learning approaches on raw time series data to further improve early prediction performance and generalization capability.